

3. (a) The micrograph below shows a human sperm cell.



5µm

(i) Calculate the actual length of the sperm cell. [2]

..... µm

(ii) The mitochondria in the sperm do not enter the secondary oocyte at fertilisation. What does this suggest about the origin of all mitochondria in an organism? [1]

.....

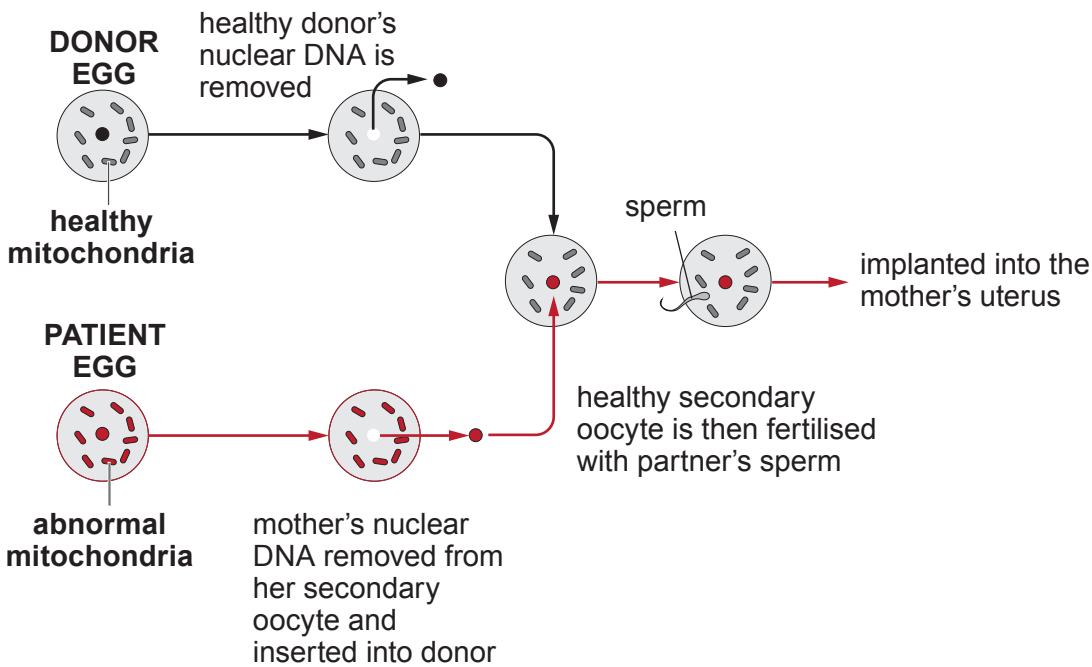
.....

(iii) A woman has inherited a mutation in her mitochondrial DNA resulting in mitochondrial disease. Her husband and father do not have this disease. Complete the table below to indicate which of her relatives will suffer from the **same** mitochondrial disease. [2]

Relative	Mitochondrial disease (✓ or x)
Son	
Daughter	
Daughter's children	
Son's children	



(iv) The inheritance of mitochondrial diseases can be prevented using the technique outlined in the following diagram.



Using all of the information provided explain why a child produced by this method has been referred to as a **‘three parent baby’**. [3]

.....

.....

.....

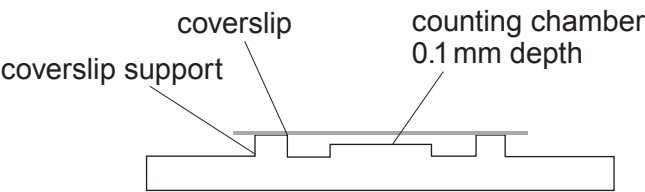
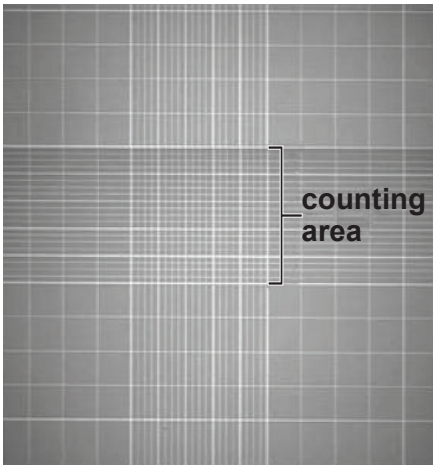
.....

.....



(b) A haemocytometer can be used to calculate the number of sperm in a sample of semen.

A haemocytometer is a microscope slide with a rectangular chamber at the centre which is engraved with a grid as shown below. A coverslip is supported over the chamber and the depth of the chamber is known. The number of sperm cells in a specific volume of semen can be counted.



The counting area is observed using a microscope and the number of sperm cells is counted. The sperm are killed before counting.

(i) Explain why this technique is not suitable for counting live sperm. [1]

.....

.....



Examiner
only

(ii) The semen sample was diluted by adding 0.1 cm^3 of semen to 9.9 cm^3 of solution. The number of sperm cells in the diluted sample in the counting area was 40. The counting area is 1 mm^2 with a depth of 0.1 mm .

I. Calculate the number of mm^3 in 1 cm^3 . [1]

number =

II. Given that in each 0.1 mm^3 there are 40 sperm cells, calculate the number of sperm cells in 1 cm^3 . [1]

number =

III. The semen sample was diluted by adding 0.1 cm^3 of semen to 9.9 cm^3 of solution. Calculate the number of sperm cells in the original 1 cm^3 of semen. [1]

number =

12

